

# AZIZULLAH KHAN

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## EDUCATION

### Institute of Business Administration (IBA)

Bachelor of Science in Computer Science **CGPA: 3.76/4.0**

Karachi, Pakistan

August 2021 – May 2025

- Awards: Dean's List (awarded for outstanding academic performance, 4 consecutive years).
- Coursework: Data Structures, Algorithms, Object-Oriented Programming, Software Architecture & Design, Computer Architecture & Assembly, Operating Systems, Parallel & Distributed Computing, Machine Learning.

## SELECTED SYSTEMS PROJECTS

### Parallel Shortest-Path Performance | Python, Multiprocessing, NetworkX

May 2025

CSE467 – Parallel & Distributed Computing

- Compared sequential and multiprocessing Dijkstra, Bellman-Ford, and A\* using matched workloads and timing.
- Measured the task-granularity threshold for Python multiprocessing, finding a crossover around **1 ms** of work per task; batching independent shortest-path queries reached **4–5×** speedup. [Preprint](#).

### SHA-256 Placement on xv6-riscv | C, RISC-V, xv6

December 2024

CSE331 – Operating Systems

- Implemented the same SHA-256 core in kernel space, user space, and through a system call to compare placement costs.
- Measured inputs from 50–420 bytes over 100 runs per size, finding that the system-call path added **2–4%** over user-space execution for inputs of 100 bytes and above. [Preprint](#).

### Indus Sahulat – Final Year Project | Spring Boot, PostgreSQL, Redis, WebSockets

September 2024 – May 2025

Supervised by [Dr. Sajjad Haider](#)

- Built a real-time emergency response platform for Indus Hospital, connecting patients, ambulance drivers, and dispatch workflows; reduced measured ambulance response time by **~10%**.
- Implemented live ambulance tracking with WebSockets and Redis, reducing peak database writes by **~20%**, with Spring Security across four user roles. [Final Report](#).

## EXPERIENCE

### Data Science Dojo

August 2025 – Present

Software Engineer I, Infrastructure

Remote

- Engineer production **AKS** infrastructure, including cluster maintenance and upgrades while maintaining availability.
- Designed a zero-trust hub-and-spoke network across **8** spokes, and built modular IaC used across **8** environments.
- Reduced environment setup time from **~2** hours to **20 minutes** through reusable infrastructure modules.
- Led an event-driven billing service spanning **8** usage dimensions across **5** tenants, with tracing for LLM requests and costs.
- Built centralized monitoring covering **95%+** of tenants, detecting production issues within **~8 minutes** and identifying **10+** issues before customer impact.

### Salsoft Technologies

January 2024 – July 2025

Junior Backend Developer

Karachi, Pakistan

- Developed backend services for **3** production apps, including reusable access-control, notification, chat, and search modules.
- Sustained p95 latency on high-traffic endpoints through database indexing, query optimization, and caching as usage grew.

### Institute of Business Administration (IBA)

August 2023 – May 2024

Teaching Assistant – Department of Computer Science

Karachi, Pakistan

- TA for **CSE141** Introduction to Programming (C++) and **CSE142** Object-Oriented Programming.
- Ran weekly labs, designed assignments and held one-on-one sessions to help students debug their code.

## TECHNICAL SKILLS

**Languages:** Python, Go, C, C++, Java, TypeScript, JavaScript

**Frameworks & Libraries:** Docker, Kubernetes, FastAPI, Spring Boot, Node.js, React, vLLM

**Developer Tools:** Azure, AWS, Terraform, Slurm, CI/CD, Linux, Prometheus, Grafana, Datadog, PostgreSQL, Redis

## LEADERSHIP

### IBA Computer Science Society

2024 – 2025

Module Head, Competitive Programming

Karachi, Pakistan

- Led the Competitive Programming module to **100+** participating teams, organizing technical challenges and activities; recognized with the **Best Module Head Award**.

### IBA Data Science Society

2023 – 2024

Module Head, Databases

Karachi, Pakistan

- Led the Databases module and coordinated a technical organizing team, developing and delivering database-focused challenges and activities for students.